China Methane Infrastructure Mapping

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Executive Summary

This analysis, conducted in collaboration with the Environmental Defense Fund (EDF), represents a significant step forward for mapping previously publicly unknown oil and gas infrastructure. The China Methane Infrastructure Mapping project is a research project conducted by the EDF that is tasked with locating oil and gas infrastructure in China that could potentially be releasing methane emissions. In order to do that they need data of oil and gas infrastructure, which is not available through official government, industry, academic, or NGO sources for the country of China. To remedy this dearth of information, this preliminary analysis was conducted to produce a dataset of over 15,000 geolocated facilities identified via Open Street Map. From that dataset satellite imagery was gathered for the Ordos region; the largest emitter of methane relative to the other oil and gas producing regions in China. This imagery dataset contains over 1,700 images of industrial facilities within the bounds of the MethaneSAT scene. A machine learning model was developed to identify tanks using annotated satellite imagery from Airbus Defence & Space. This model was then run on that gathered satellite imagery and resulted in a dataset of over 3,000 positively identified tanks in the Ordos region. The data from every step of this process and code used to generate that data are all included in the attached GitHub repository (appendix B).

Introduction

This project, conducted to support the Environmental Defense Fund, an international non-profit that uses science and economics to address environmental issues, locates oil and gas infrastructure in China as possible point source emissions in areas with a high concentration of oil and gas captured by MethaneSAT scenes. MethaneSAT is a satellite launched with the goal of monitoring methane emissions from space, with a focus on finding both smaller sources, and area-wide emissions of whole oil and gas basins, previously impossible to do with other atmospheric gas tracking satellites. As part of this effort, the EDF has been tracking oil and gas infrastructure in order to attribute the methane emissions tracked by the satellite to specific facilities on the ground. The EDF is interested in supplementing their Oil and Gas Infrastructure Mapping (OGIM) dataset with additional data sources. Currently the OGIM dataset, which does not include any data from China, is the “result of extensive efforts at EDF and MethaneSAT, involving the acquisition, analysis, curation, integration, and quality assurance of public-domain datasets sourced from official government reports, industry publications, academic studies, and various non-government entities” (Roy, n.d.).

The oil and gas infrastructure datasets of this type are not publicly available for China. Because of the lack of publicly available datasets on this topic, the EDF is interested in utilizing other sources of data in order to map the infrastructure. For the purposes of this internship the other data sources include Open Street Map, heads up digitization from Bing Imagery and machine learning. These techniques were able to identify nearly 15,000 thousand suspected oil and gas facilities across 12 target MethaneSat scenes in China. In addition 3,390 new tanks were identified in the Ordos basin alone with machine learning techniques. In addition to the actual numbers of facilities and tanks, in identifying the infrastructure, we developed a process that can easily be reproduced for other countries and easily expanded upon to map the rest of China’s methane infrastructure. This represents a large step forwards in the global infrastructure mapping project.

Background

Why do we care about methane?

Methane, like all other greenhouse gases, traps heat in our atmosphere and contributes to global warming. However, methane is notable in the hierarchy of greenhouse gases for its exceptional ability to trap heat while remaining relatively short-lived. According to the EPA methane is currently “accounting for about 11 percent of global emissions. Methane is more than 28 times as potent as carbon dioxide at trapping heat in the atmosphere” (Importance of Methane | US EPA, 2026). However, despite the effectiveness of methane at trapping heat in our atmosphere it also decays into CO_2 after an average of 9 years (Karakurt, Aydin, & Aydiner, 2012). This makes cutting down methane emissions an effective short term target for lowering the amount of greenhouse gases trapping heat in our atmosphere. In addition, methane is frequently emitted as a byproduct of other industrial activity (Karakurt, Aydin, &

Aydiner, 2012). This is due to certain sectors being unaware of the amount of methane emissions they are producing, both from ignorance and a lack of enforcement of effective regulations. From these facts it is clear why the EDF's chief scientist and MethaneSAT project leader explained “some call it low hanging fruit. I like to call it fruit lying on the ground” (Carrington, 2024). Properly identifying and tracking the facilities and the owners of those facilities producing methane through projects like MethaneSAT can both “name and shame” emitters as well as “document progress that leading companies are making in reducing their emissions” according to Mark Brownstein, the EDF's senior vice president (Carrington, 2024).

MethaneSAT

The EDF in partnership with the New Zealand Space Agency along with funding from the Bezos Earth Fund, launched a satellite, MethaneSAT, in order to monitor methane emissions across the globe (Kuthunur, 2024). MethaneSAT is a high precision, wide view path satellite designed to track methane emissions at various oil basins around the globe over time. Oil basins are targeted as these geologic features are where fossil fuels develop under the ground and therefore oil and gas infrastructure is likely to be located right on top of the oil. As part of the project of tracking global methane emissions, the goal of the MethaneSAT is to “identify and quantify total emissions from small sources across wide geographic areas -- not just the large, so-called “super emitters” that can be seen by other satellites. This valuable addition to existing research changed the scientific and policy conversations by definitively showing the huge role diffuse sources play around the world” (MethaneSAT, 2025). This size and scope of this project is unique due to the satellite's ability to track both smaller point source emissions as well as area wide emissions. While it is not the first satellite able to do either of those things, it is the first to do both at the same time from the same satellite. This combination is able to gather much more information about the sources of pollution at a far more rapid timescale compared to previous methods of relying on two separate satellites with different orbits.

The satellite data is separated into scenes covering all or part of the many oil and gas basins across the world. These areas have a high concentration of oil and gas infrastructure due to the fact that this is where the oil is extracted from the ground and account for 80% or more of the oil and gas produced on earth (MethaneSAT, 2025). Each MethaneSat scene contains “heatmaps of 1 km^2 areas across targets that are 200 km x 200 km, with a native pixel size of 100m x 400m” (MethaneSAT, 2025). Once the data from the satellite has been received, cleaned, and processed it is then published to the public for free via MethaneSAT’s web portal and Google Earth Engine (MethaneSAT, 2025). The goal of MethaneSAT’s radical transparency is to help identify and fix the emissions without the need to spend money or have access through an institution. Due to the massive adverse impact of greenhouse gas emissions, any step to make finding the solutions more accessible is a step in the right direction.

A goal of the project involved attribution to specific facilities, tanks, and pumpjacks, the mechanism used to pump oil from underground sources. To highlight the importance of this attribution step, an EDF study with MethaneSAT data showed that in the Permian Basin, the largest oil and gas basin in the world, the emissions intensity on the Texas side of the basin was more than double of the intensity on the New Mexico side. Despite the geography and infrastructure being approximately analogous across the border. This highlights how effective New Mexico’s more stringent emissions reduction strategies are at achieving their goals (MethaneSAT, 2025). Going beyond the simple naming and shaming, this methodology can uplift effective strategies or point out where those strategies fall short.

Loss of Contact with MethaneSAT

On Friday June 20, 2025 MethaneSAT lost power and subsequently lost contact with earth. The loss of the satellite does not mean the end of this project as there is still more than a year's worth of data to be processed and analyzed. The mission itself was a success on all fronts, save for longevity of the satellite (MethaneSAT, 2025). The spectrometers performed exceptionally well and the insight into the amount and distribution of methane emissions show the full scope of the problem and the most viable areas for effective methane emissions reduction. As of May 2026 there are no public plans for relaunching a replacement satellite.

MethaneSAT in China

This type of analysis is made possible through the precision of MethaneSAT and the availability of data on exactly where the oil and gas infrastructure is located. Using the precise locations of infrastructure along with advanced modeling in order to trace emissions back to their source, it is possible to attribute those emissions to specific facilities. In many countries this data is freely available, however, with China that data is not made public. As such the goal of this project is to use satellite imagery in order to pinpoint possible point source emissions for the target regions in China, which include the Ordos Basin, the Songliao Basin, Bohai Basin (North China), Junggar Basin and Tarim Basin.

The main infrastructure targets of this are oil and natural gas processing facilities, which contain oil pumpjacks and storage tanks for oil and gas. Each of these pieces is a step in the oil extraction pipeline and has the potential to allow for unintended methane emissions. Due to the physical and geological differences of each of the target basins, the infrastructure will be slightly different across each of them according to need and the technology available when each basin was developed. As such, there is no one size fits all approach to identifying the infrastructure in each and they must be treated on a case by case basis in order to assess each as accurately as possible.

Study Areas

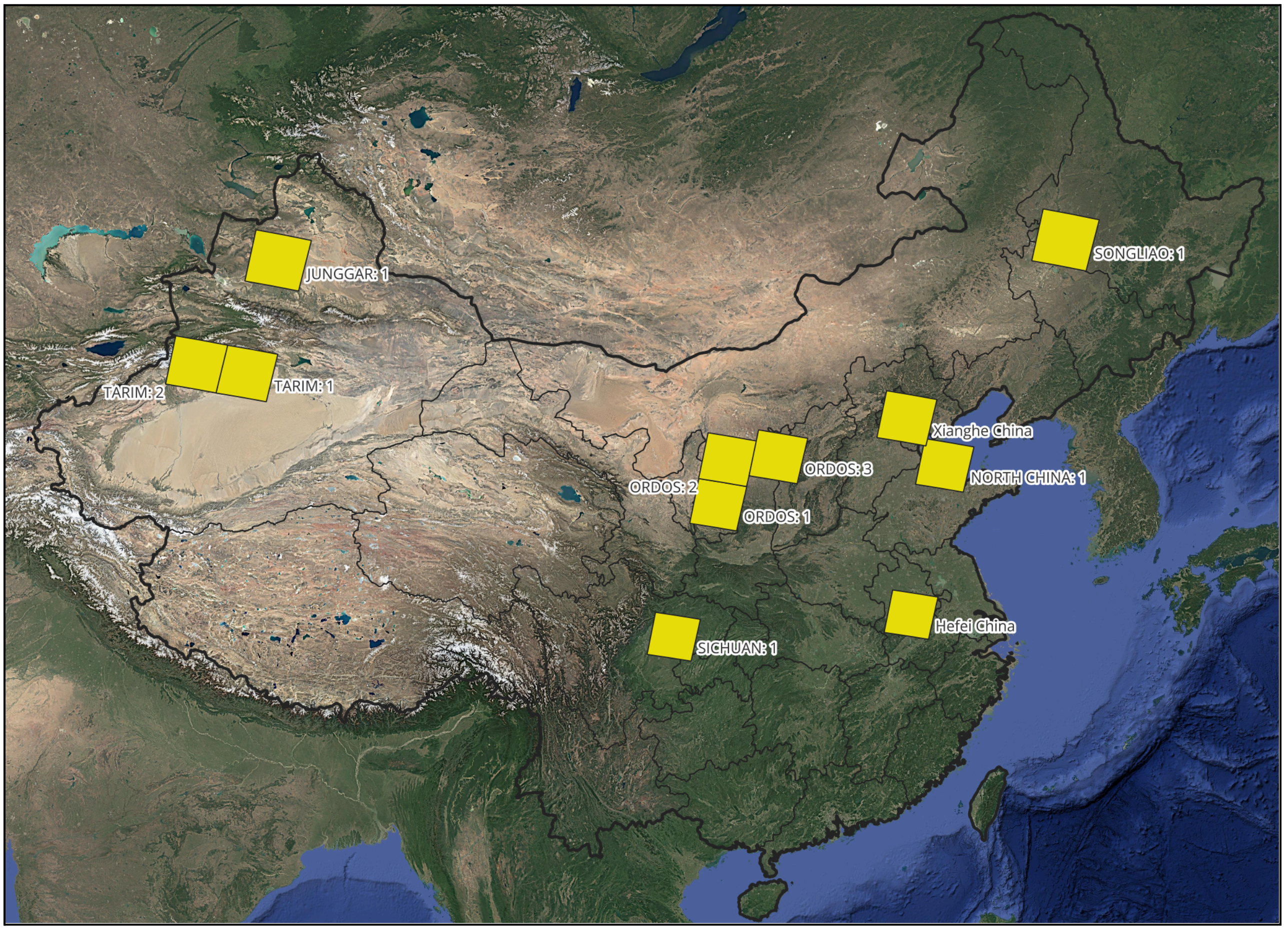

Figure 1: Published MethaneSAT Scenes in China

The areas of greatest interest for China EDF captured by MethaneSAT are: the Ordos Basin, Songliao, Bohai (North China) Basin, Tarim Basin and the Junggar Basin. For tank identification the current scope of the project was limited to the Ordos Basin. While these are not all of the basins covered by MethaneSAT scenes in China, they are the ones of most interest due to a combination of the physical nature of the infrastructure there as well as the total emissions produced in these target areas.

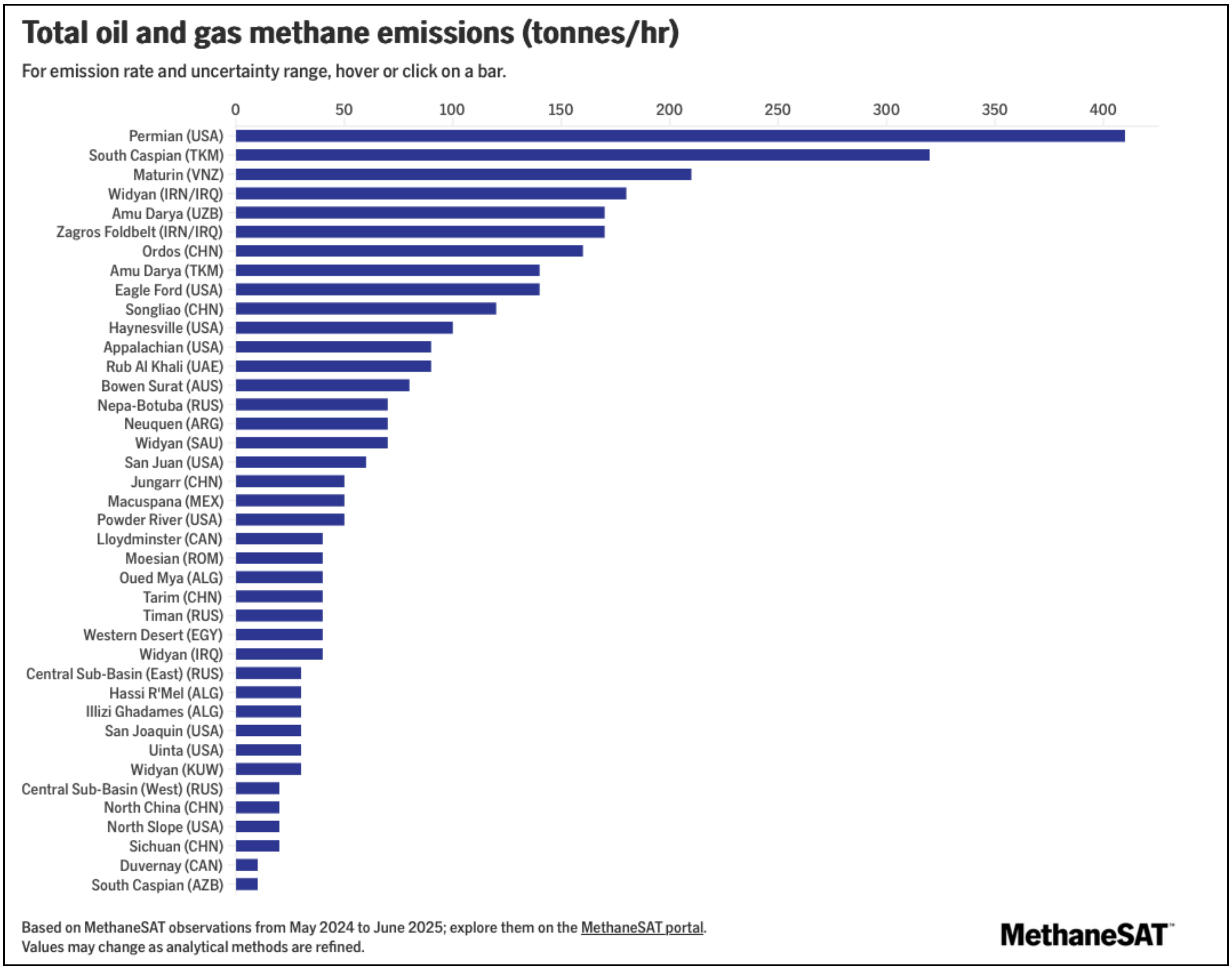

Figure 2 : Total Oil & Gas Methane Emissions, (MethaneSAT, 2025a)

As shown in the graph at Figure 2 based on data from MethaneSAT, Ordos and Songliao are the two highest methane emitters of the basins in China in tonnes per hour. Those two basins rank 7th and 10th in the world.

Ordos Basin

The Ordos Basin is the main focus of the research as it is the largest emitter of methane gas in China according to the MethaneSat data. The basin itself is considered a "considered a super petroliferous basin" as it contains "several superimposed petroleum systems and a proven reserve of 200 billion bbl of oil and 160 TCF of gas" (Liu et al., 2023). This and its relatively recent development compared to the other Chinese basins bring it to the top of the

EDF’s list of priorities.This area is home to both traditional coalbed mining as well as shale gas. It is covered by a total of three published MethaneSAT scenes. Geographically in China, the Ordos Basin spans the Shaanxi province (陕西), Gansu province（甘肃), Shanxi province (山西), Ningxia Autonomous Region (宁夏回族自治区), Inner and Mongolia Autonomous Region(内蒙古自治区).

Other Basins

Songliao Basin is located in the northeast part of China covering the Heilongjiang, Jilin, and Liaoning provinces and touching the Inner Mongolia Autonomous region as well. It is of interest due to the oil pumpjacks located in this area. Another potential emitter of methane and a possible future candidate for identification with machine learning analysis.

The remaining basins in China are of note as they represent the later phases of research. However, for the time being the precise geologic and geographic details are not of great importance. The Junggar Basin is home to large supplies of traditional coalbed mining (Lee, 1985). The North China region comprises several separate on and off shore oil and gas operations. Due to the proximity of this basin to both Beijing and a large population, it has a long history of development. Most recently “game-changing discovery has been made in China with the successful completion and production of terrestrial shale reservoirs, which have maintained a natural flow for more than 260 days” (Pu, 2019). This recent development and success marks a step forward for Chinese deep shale exploration. The success of this project is noteworthy as increased oil and gas production can lead to an increase in methane emissions.

Methods

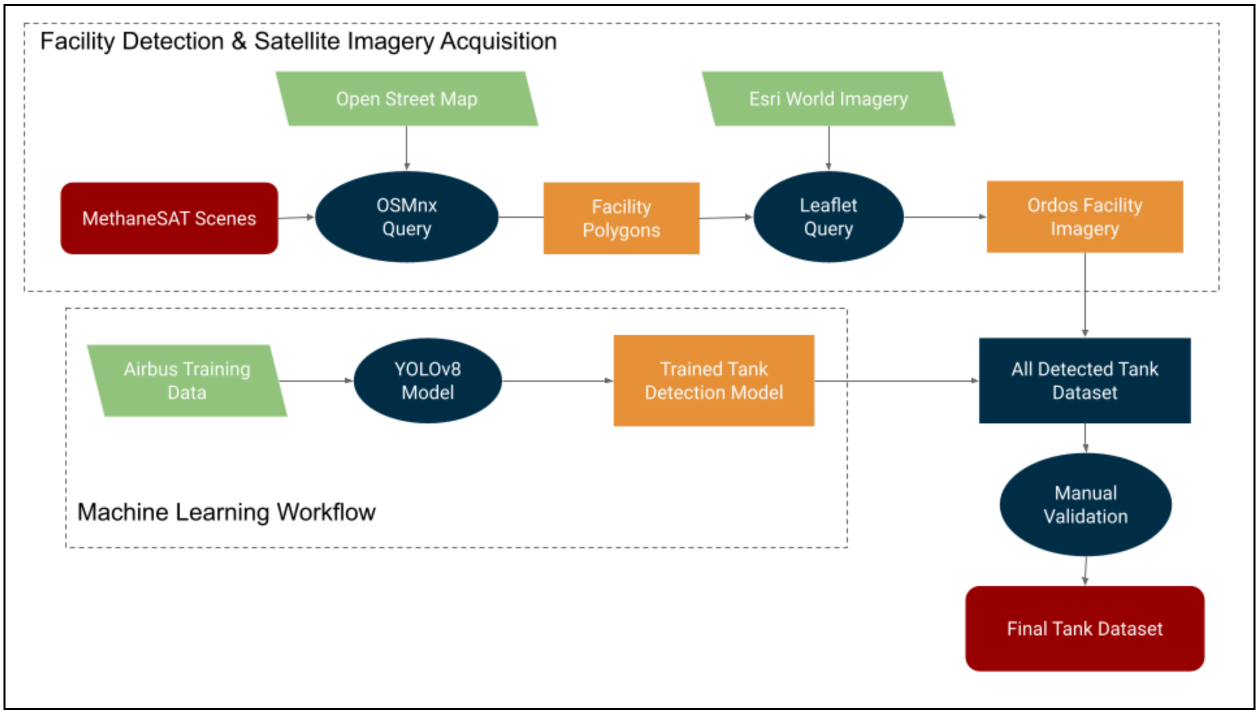

Figure 3: Combined Workflow Diagram

A) OSM and Facilities Polygon Query

The main source of information for identifying facilities came from Open Street Map (OSM). Open Street Map, founded in 2004, is leading the charge to make geographical data free and accessible to the public (OpenStreetMap Wiki, n.d.). To do this Open Street Map has crowdsourced the massive project of mapping the entire globe. One of Open Street Map's main goals is to create "a free dataset which will enable programmers, citizen scientists, social activists, cartographers and the like to fulfill their plans without being limited" (OpenStreetMap Wiki, n.d.). Due to the nature of all data and specifically crowdsourced data, the coverage, completeness, and accuracy is not perfect, however, neither is that of proprietary mapping services.

While it is imperfect, it is a starting point for this project. In order to access Overpass Turbo, OSM's public API for accessing their mapping data, OSMnx, a Python package developed for this exact task was utilized (Boeing, 2025). In order to locate facilities, the OSM specific tags identifying those specific buildings must be used. In this case the tags of industrial buildings are as follows:

```
'building': 'industrial',
'landuse':'industrial',
'man_made':'works'
```

Using the MethaneSAT scenes as the search area, `features_from_polygon` function from OSMnx was queried. This function searched the target polygon, in this case the area of the MethaneSAT scene, for features matching the target tags. Using this function we are able to get all of the features in the Open Street Maps database matching the 'industrial' or 'works' category. This serves as a proxy for all of the industrial buildings in the target areas. This function returns a Geopandas Geodataframe, a two dimensional datatype with a column representing each features geography, similar to a spreadsheet except it contains geographic information. This geodataframe and the polygons within it are one of the final deliverables for this project, as it is the first step towards mapping oil and gas infrastructure in the target regions. The geodataframe was further cleaned in Python.

The first thing calculated was the area of each facility, as the size of each facility can help identify what specific industrial processes and machinery are located in that area. The next processing step was to calculate the centroid of each facility and return its latitude and longitude. This is important as the geometry field on the geodataframe is not easily human readable, therefore, converting that complex field into simpler latitude and longitude columns helps increase the accessibility of the data to more people. The files were then saved as a geopackage, an open and platform independent file type for storing geospatial data (Admin_Ogc, 2026), as well as a standard version saved as a spreadsheet for easier access for a non-geospatial audience. The programmatic nature of this query means it is easily replicable across all the MethaneSAT scenes, and any arbitrary geographic region. The full code is included in the GitHub repository in appendix B as a Python file named `OSM_query-step1.ipynb.`

B) Leaflet and Imagery Query

While the geospatial information gathered from the Open Street Map query is useful in identifying the facilities, it does not cover what is in those facilities. An additional target is the oil and gas storage tanks located on those facilities, so satellite imagery was required for those facilities. Due to both the intensive nature of this particular task and the priority given to Ordos, it was only done on the Ordos basin, specifically the Ordos 1, 2, and 3 MethaneSAT scenes. To gather satellite imagery another query was generated, this time using the leafmap package, an open source package built of leaflet, a lightweight JavaScript library for interactive mapping (Leafletjs n.d.). Because high quality satellite imagery can create large files very quickly, the targeting for this query had to be very specific. Since we know that oil and gas storage tanks are likely to be on oil and gas facilities, we can use the facilities geodataframe generated from the OSM query as a starting point. The first technique used to reduce the processing is to dissolve all the polygons. This is only useful for facilities or subfacilities that contain overlap. The dissolve simplifies the geospatial feature by aggregating facilities that overlap. See Figure 4 below for a visual example of the dissolve.

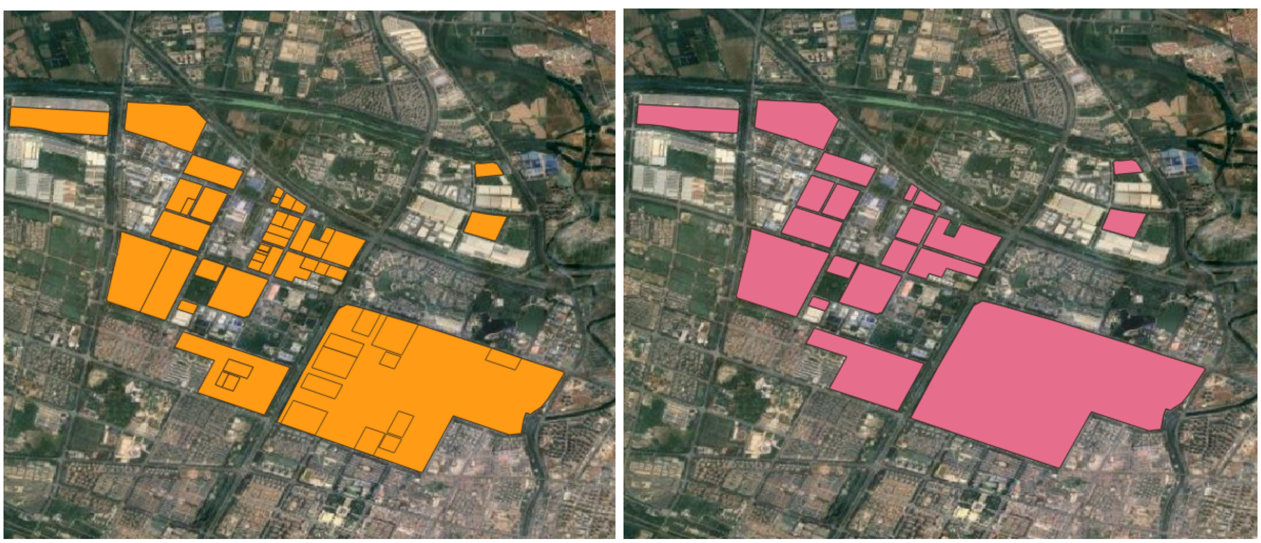

Figure 4: Example of the dissolve function aggregating overlapping facilities

Using the dissolved polygons of facilities generated from the last query a minimum bounding box was created around them. This leaves a new set of polygons, each a uniform rectangle, entirely enclosing the original facility as shown in Figure 5 below.

Figure 5: Example Facility (Blue) with Minimum Bounding Box (Red)

This bounding box is generated for each facility and is the basis for the imagery query; it creates a standard rectangle covering the entire area of the facility. One final step in filtering the facility data is to eliminate any facility bounding box with an area greater than 8km^2 as that is the limit of the size of imagery able to be queried. With a new geodataframe containing the bounding boxes, a query is run on the leaflet package for ESRI World Imagery, a mosaic of high quality satellite imagery published by ESRI and available for use for non-commercial uses (ESRI | ESRI Legal, n.d.). This loops through the bounding boxes and produces a high quality satellite image in the form of GeoTIFF, a geospatially located raster image, with a spatial resolution of approximately 1.2 meters per pixel. This imagery was queried one MethaneSAT scene at a time and took approximately 7 seconds per facility to complete due to rate limits on the amount of queries placed and download speed. This Ordos facility imagery dataset will be utilized for the machine learning tank detection. The code for this step is included in appendix B as `imagery_acquisition-step2.ipynb`.

C) Machine Learning Tank Detection

The first step for machine learning is finding a dataset. For this we used the Airbus Oil Storage Detection Dataset. Airbus Space and Defence is the military and space division of Airbus, the aeronautics company (Airbus Defence and Space - Space Digital, n.d.). They have provided an annotated satellite imagery dataset of 103 oil storage tanks around the world as a sample of the imagery offered by their OneAtlas service.

The next step in the process is to split the data into training, validation, and testing subsets. The training data set is used for the first stage of machine learning. In this stage the model “learns” the data, tuning various internal parameters, like neural network weights, to try to accurately predict where oil storage tanks are located. The validation data is then used to prevent overfitting and further tune the model. Overfitting occurs when a model has been trained too closely on the training dataset and it does not perform well on new data it has never seen before, analogous to a student who has memorized the practice questions only to fail the test because the teacher asked different questions. The validation dataset is used to check whether the model is overfitted to the training data and then it tweaks other hyperparameters to see what performs the best. Lastly, the testing dataset is used as a completely unbiased and unseen final check of the model’s performance. The model has not seen this data before and it can be used to check the accuracy of the model’s performance on unseen data.

For the base model we used YOLOv8 from Ultralytics. Standing for “You Only Look Once”, YOLOv8 is a computer vision model designed for object detection, segmentation, classification and other computer vision applications (YOLOV8: State-of-the-Art Computer Vision Model, n.d.). While there were only 103 images in the dataset, among those images there was a total of 13,593 individual tanks. The model was set to run with 100 epochs, meaning it was trained on each image 100 times while tuning the weights to try to be as accurate as possible. The successful run produced a trained model for detecting oil storage tanks from satellite images.

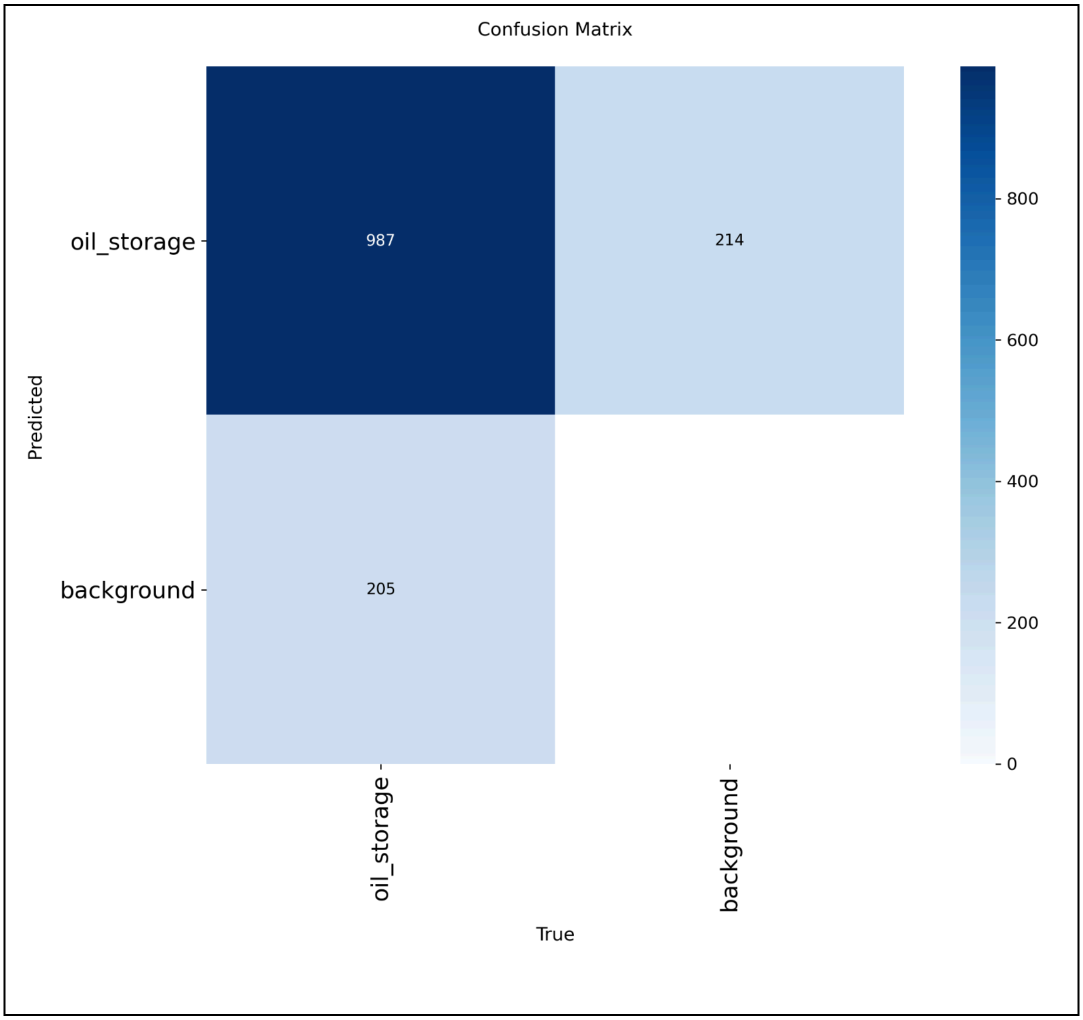

Figure 6: Confusion Matrix

Testing the trained model on the test dataset results in a confusion matrix, shown in Figure 6 above, which is a matrix representing the accuracy of the model on the test images that it has never seen. Interpreting the graph we have the x axis representing the actual values and the y axis representing what the model predicted. The top left quadrant represents oil storage tanks that were accurately predicted. The top right quadrant represents false positives, where the model incorrectly identified an oil storage tank, but there was not one located there. The bottom left quadrant is false negatives, where the model did not predict an oil storage tank despite one located there. Lastly the bottom right quadrant is an artifact of the design and it is not relevant to the analysis, it is technically where the model predicted background and there was background. This resulted in 987 correct detections, 214 false positives, and 205 false negatives, for an overall accuracy of 73%. However, since the final product will be up for manual review a more applicable statistic for the model is that it identified 83% of the tanks correctly. This statistic ignores the false positives, as those will be screened out during the manual review process. The code is included in appendix B, `yolov8_step3.ipynb`.

D) Machine Learning on Gathered Imagery

Once the imagery for the oil and gas facilities and a trained model for detecting tanks within those images is available, the next step is to start identifying tanks in the target area. Running the model on the Ordos facility imagery obtained in section B, an initial detected tank dataset is generated. Due to the nature of geospatial data and image sizes, the output from the YOLO model is a text file containing the x,y coordinates of the tanks in each of the images. Therefore, further manipulation of the dataset is required to determine the real-world location of the tanks. Since the images are already geolocated, some simple arithmetic operations are needed to turn those pixel coordinates into actual coordinates. This math is also in the `yolov8_step3.ipynb`. After the processing we are left with the geolocated predictions of tanks in the target regions.

The final step in the process is to manually validate the predicted dataset. To do this each identified tank must be hand checked and validated for accuracy. The QGIS Plugin GoToNextFeature3+ was utilized to quickly iterate through the dataset and leave flags marking the status of each detected feature. The schema is as follows: correctly validated images are marked as no error, type I errors (false positives) are flagged as incorrect, and the rest are further marked for more intensive human review. This captures type II errors (false negatives) if they can be seen within the image, as well as tanks that were identified twice or identified in the wrong location. It lastly includes the several tanks that were identified multiple times as they are on the border of the two MethaneSAT scenes and the imagery was duplicated. Examples of each of these images can be seen in the Appendix A.

The model detected 4,188 total tanks. Of those identified, 3,028 were correctly identified (72%), 714 were false positives (17%), 394 were flagged for further corrections, missing adjacent tanks or double counted (9%), and 52 were on borders between scenes (1%). Since there is no way to confirm how many tanks were missed without manual identification on all of the imagery, a full picture of the accuracy of this model on the Ordos data cannot be achieved at the moment. However, the results do appear to be in line with the results on the test dataset as the oil and gas storage tanks do not significantly change visually from country to country. Given that information, this is a very promising result and a great starting point for oil and gas tank identification.

Results

Facility Polygons and Imagery

The results from the Open Street Map query and the manual digitization are set forth in the table below. The final dataset contains over 15,000 suspected oil and gas facilities, with slightly over half of the facilities being explicitly named. With those names we can further investigate whether they are in fact oil and gas facilities, as well as trace the ownership of those facilities to further improve the value and usability of the dataset. This is a large step towards creating a spatially explicit database of all oil and gas infrastructure across China. This data is made available and included in the supplemental data folder as both an Excel spreadsheet and a geopackage. The geopackage is meant for users interested in mapping or further manipulating the spatial data, while the Excel sheet is there for further aspatial analysis. An important note for anyone doing further analysis is that not all csv writing functions natively handle Chinese characters. This is why the output is available as an Excel spreadsheet, so caution is warranted if writing this data to a csv.

Location	Count	Sum of Area (km^2)	Named Facilities
Junggar	320	88.2	7
North China	11,236	2,130.7	6,522
Ordos	2,660	618.7	1,120
Songliao	100	5.4	0
Tarim	861	181.7	30
Total	**15,177**	**3,025**	**7,679**

Table 1: Summary Statistics of Facility Polygon Dataset

The analysis shows that the North China scenes have by far the most facilities, as well as the most named facilities. This could be due to the scene's proximity to Beijing and the area's high population. Since OSM is a crowd sourced platform, having a larger crowd could contribute to the completeness and level of detail in the entries. In addition to the located facilities, for the Ordos basin we have a dataset of high quality satellite imagery. For this region alone we have 1,741 geolocated images of facilities. While in this instance they were used for the machine learning process, they are also a useful snapshot of the identified facilities and could be used for other related analysis.

Tanks Detected with Machine Learning: Ordos Basin

Running the trained YOLOv8 model on the images acquired for the Ordos imagery and processing the initial results results in a dataset consisting of 2,983 correctly identified tanks and 407 flagged for further review. While these tanks were initially identified with a bounding box around them, the processing involved using the minimum side length as the diameter to generate a circle around the centroid of the rectangle. The result is a circle approximately covering the shape. This is useful both as a visualization tool, and because the area of that circle can be used to approximate the size and volume of oil stored in that tank. This produced another useful piece of information to get an accurate assessment of the oil and gas infrastructure in the target area.

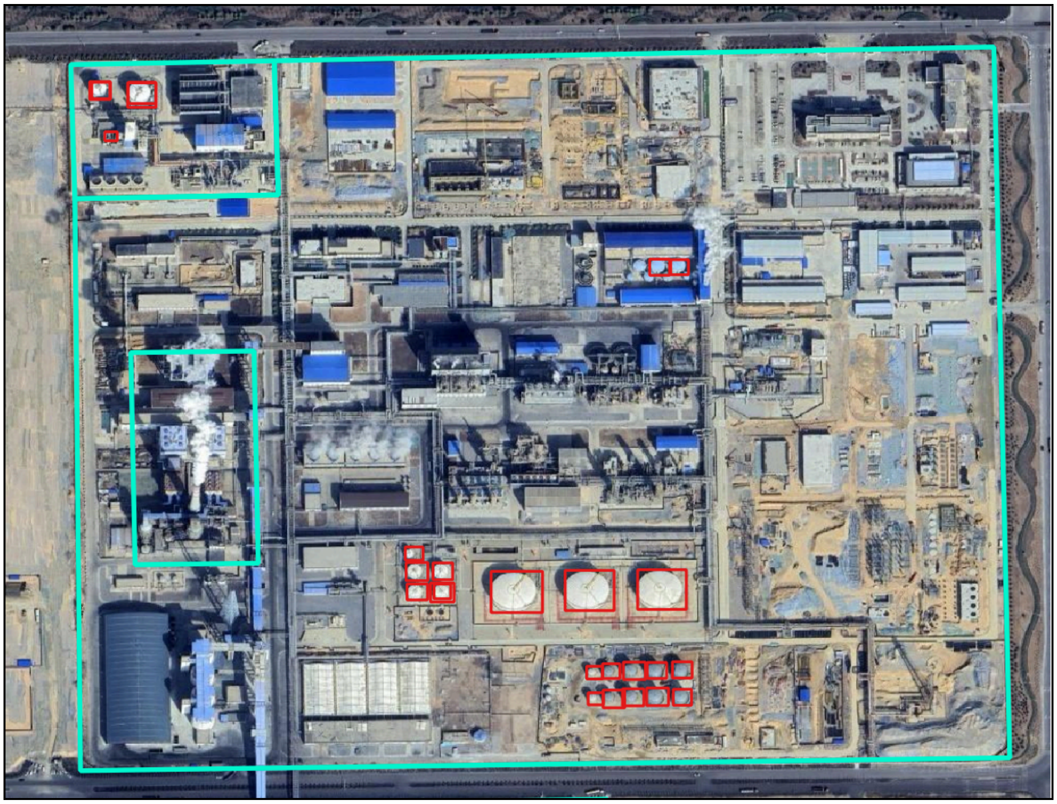

Figure 7: Facility with Detected Tanks

Workflow and Reproducible Code

In addition to the specific infrastructure identified, the project also successfully created a functional code and workflow process. This code is available in the supplementary data GitHib repository, and will be further improved and iterated on by the researchers continuing to work on this project. This first step was a proof of concept and affirmation that this work is possible and can be done within the constraints of the organization.

The value of the code and workflow developed for this project extends beyond China. Since the underlying oil and gas infrastructure does not vary greatly depending on the country, this type of work can be done in other countries that do not make their oil and gas infrastructure data publicly available. It can also be used in countries with incomplete infrastructure inventories.

Conclusion

Project Limitations and Possible Continuations

While this project has been an extremely successful exploratory analysis of mapping oil and gas infrastructure in an area without public data sources,the time constraints imposed by the internship made limited additional work which would have improved the results. The first component that could be improved is the almost complete reliance on Open Street Map data. While OSM is convenient and generally reliable, it is not best practice to only use crowd-sourced data without any other forms of verification, whether that is any other database or simply manual doublechecks. In addition to just confirming the accuracy of the OSM data, a further check of the facilities identified could be useful in determining what specific type of facility is located there. This effort would require some expertise in oil and gas manufacturing, and the ability to easily discern the types of facilities through satellite imagery, but it would be a tremendous addition to the project. In addition, cross-referencing the named OSM facilities] with existing data sources could further identify facility type and ownership. Finally, some Chinese language sources, such as Baidu could be used for this crossreferencing.

With regard to the machine learning component of the project, while the currently trained model is accurate, it could be further refined. Taking the examples from the Ordos region, both the positive and negative ones, and using those to further train the model would be an effective way to further improve accuracy when the model is used on the unstudied basins in China. In addition, the flags from the first pass of the manual check should be checked and resolved before further analysis.

Locating additional sources of imagery and running the object detection model could be another way of locating tanks and accounting for potentially missing tanks. These are possible solutions, but in order to be thorough, a manual check of all images should be conducted to determine if there are any tanks that were not captured by the object detection model. Without the full manual verification, it is impossible to know the full accuracy of the model and it would be wise to do that before spending time using the model on the other Chinese Basins.

With this machine learning workflow, pumpjacks, another component of the oil production system and a potential methane emitter, could be identified using satellite imagery. These are mostly located in the Songliao Basin and its related MethaneSAT scenes.

Pumpjacks are a potential identification target and would be a worthwhile type of oil and gas infrastructure to be identified. The main impediment to pumpjack identification is their small size and need for very high resolution imagery to accurately capture them.

End Result

The project was a success and produced the first components of a dataset to map oil and gas infrastructure in China. The first piece of the dataset is the 15,000+ facilities identified via Open Street Map. As discussed above, this dataset could be further improved with manual identification and further training from that dataset satellite imagery that was gathered for the Ordos region. This imagery dataset in addition to the Python code used to generate it are available in the attached GitHub repository (appendix B) and can be utilized by future researchers looking to gather satellite imagery for related projects.

Other products resulting from this research, the machine learning code and trained model, are also included in the GitHub repository. The model has proven to be a viable route for identifying tanks and has potential to be further refined and used for the remainder of the MethaneSAT scenes in China and across the world.

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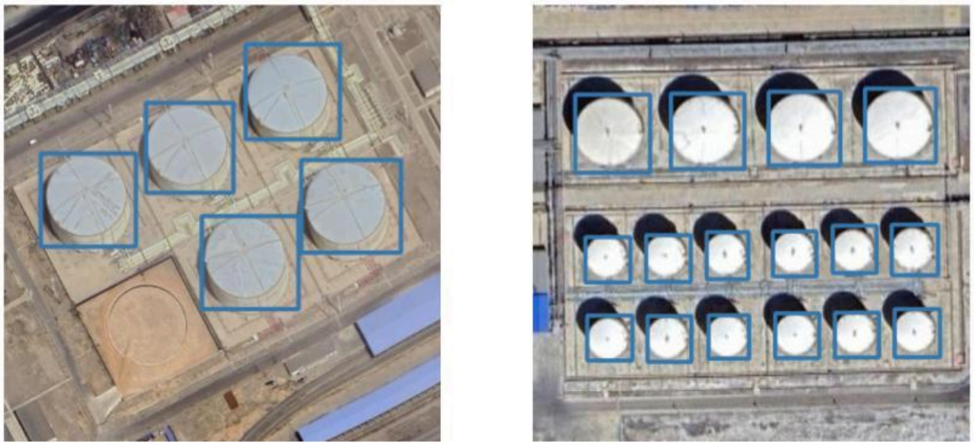
Appendix A
Schema for Manual Verification of Machine Learning Results

Validation Schema - No Error



Validation Schema - Error

Validation Schema - Flagged to fix

Appendix B
Links to GitHub Repository

https://github.com/MikeChirico/China-Methane-Infrastructure-Mapping